Eni Berilaj

Helsinki, Finland | linkedin.com/in/EniBerilaj | github.com/eniberilaj | Portfolio: Simulation engines

Profile

Computational Physicist and Mathematician bridging theoretical physics with high-fidelity software engineering. Experienced in reactor core modeling, actuarial mathematics, and building zero-dependency numerical engines from first principles. Passionate about transforming complex mathematical models—such as partial differential equations, Physics-Informed Neural Networks (PINNs), and stochastic simulations—into robust, interactive scientific software.

Education

University of Jyväskylä, Master of Science in Theoretical Nuclear Physics — 2025 – Present
Major: Theoretical Nuclear Physics

LUT-University, Master of Science in Computational Engineering and Technical Physics — 2020 – 2025
Major: Applied Mathematics

Work Experience

Mathematician, Pohjola Vakuutus Oy | OP Financial Group — 3/2025 – Present

- Develop and automate numerical and statistical models in Python (Pandas, Plotly), SAS, and SQL for large-scale actuarial pricing and reserving analysis.
- Design structured computational workflows and document advanced mathematical methodologies for risk forecasting.
- Build highly analytical dashboards and mapping scripts to handle complex financial dataset transformations.

Thesis Student, Nuclear Fuel & Core Calculations Team, TVO | Helsinki Headquarters — 6–8/2023, 7/2024 – 2/2025

- Completed Master's thesis with highest distinction, specializing purely in reactor physics rather than generalized asset analysis.
- Developed numerical models for neutron flux fluctuations and radial reflector effects in pressurized water reactors.
- Performed simulation, statistical analysis, and verification of mathematical modeling approaches in a highly regulated reactor physics environment.

Refuelling Supervisor, TVO | Olkiluoto Nuclear Power Plant — 3–5/2023, 4/2024 – 6/2024

- Updated the reactor heart map and ensured absolute operational accuracy.
- Documented and communicated refuelling activities and technical outcomes strictly to engineering stakeholders.

Radiation Protection Technician, TVO, Fortum, Alaratech Oy | Nuclear Power Plants — 4–6/2022, 8–9/2023, 1–3/2024

- Ensured radiation safety compliance and documented rigorous metrics in nuclear environments.

Selected Scientific Projects

Scientific Systems Lab — Live Web Portfolio

- **Reactor Physics Engine:** Engineered a live-solving nuclear core model featuring two-group neutron diffusion on a 45x45 mesh, six-group point kinetics, and thermal-hydraulics using custom RK4 solvers.
- **Scientific ML & PINNs Lab:** Built a manual-backpropagation Multilayer Perceptron (MLP) engine in pure NumPy. Developed Physics-Informed Neural Networks (PINNs) utilizing custom PDE loss functions to approximate the neutron diffusion equation.

- **Aerodynamics FlowLab:** Developed a real-time 3D computational fluid dynamics (CFD) wind tunnel environment using Three.js and WebGL. Features a potential-flow wake model with live stream particles, dynamic pressure mapping, and the capability to upload and analyze custom vehicle CAD models.
- **Zero-Dependency Architecture:** Designed a full-stack numerical suite relying exclusively on the Python standard library and fundamental array operations, demonstrating an elite, first-principles grasp of computational mathematics.

Skills

Languages & Core Tech: Python (NumPy, SciPy, Pandas, Plotly), C, C++, SQL, SAS, LaTeX

Scientific Computing: Numerical Methods (RK4, Finite Difference), Machine Learning (PINNs, Custom Autograd, Adam Optimizer), Mathematical Optimization (Nelder-Mead), Stochastic Actuarial Modeling

Domain Expertise: Reactor Physics (SIMULATE5, CASMO5), Core Hydraulics, Point Kinetics, Computational Fluid Dynamics (WebGL/Three.js)

Languages
Finnish: Native | English: Fluent | Albanian: Fluent | Swedish: Good